Course: Introduction to Artificial Intelligence
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MapOut – AI-Powered Travel Planner

Week 4: Uninformed Search and Representation

This week's planned objective

Define MapOut as a search problem, explore BFS and DFS algorithms, and prepare the initial attraction dataset and Python environment.

What I completed

I defined MapOut's search problem, including the initial state, state space, actions, transition model, goal test, and path cost. I also studied BFS and DFS and explained how they can explore different itinerary combinations.

For the practical part, I created a CSV dataset containing 15 attractions in Ohrid and the surrounding area, including categories, coordinates, estimated costs, visit durations, availability, and sources. I developed a Python program to load, validate, and display the data, and tested how it handles invalid or missing values.

Evidence submitted

- Week 4 report with problem representation, BFS, DFS, and search tree example.
- *attractions.csv* with 15 attractions and their details.
- *main.py* for data loading and validation.
- *README.md* with project documentation.
- Screenshots of successful execution and validation testing.

What I can demonstrate now

I can explain how MapOut represents a travel-planning problem and how BFS and DFS explore possible itineraries. I can also run the Python program to demonstrate how attraction data is loaded, validated, and displayed, including how the program detects invalid coordinates and handles unknown costs.

Problem or failure encountered

One challenge was defining how to represent different attraction combinations while considering the traveler's budget and available time. I addressed this by using a simple state representation and starting with a limited number of attractions.

Another challenge was finding reliable admission prices and opening hours. I decided to leave unverified costs empty instead of assuming they were free and documented the available sources.

Decision made this week

I decided to focus on 15 attractions in Ohrid and the surrounding area to keep the initial dataset manageable. I chose CSV for storing the data and Python's built-in `csv` module for loading and validating it before implementing itinerary-planning algorithms.

AI assistance used

I used ChatGPT (OpenAI) to clarify concepts, review examples, research attraction information, and improve my documentation. I also used it for guidance while developing my Python code. I reviewed the suggestions, implemented and tested the code myself, and made the final decisions about the project.

Feedback received

To be completed after receiving feedback.

Next week's objective

Implement and test BFS and DFS using the Ohrid attraction dataset, then explore informed search algorithms and heuristics to improve itinerary planning based on user preferences, budget, and travel distance.